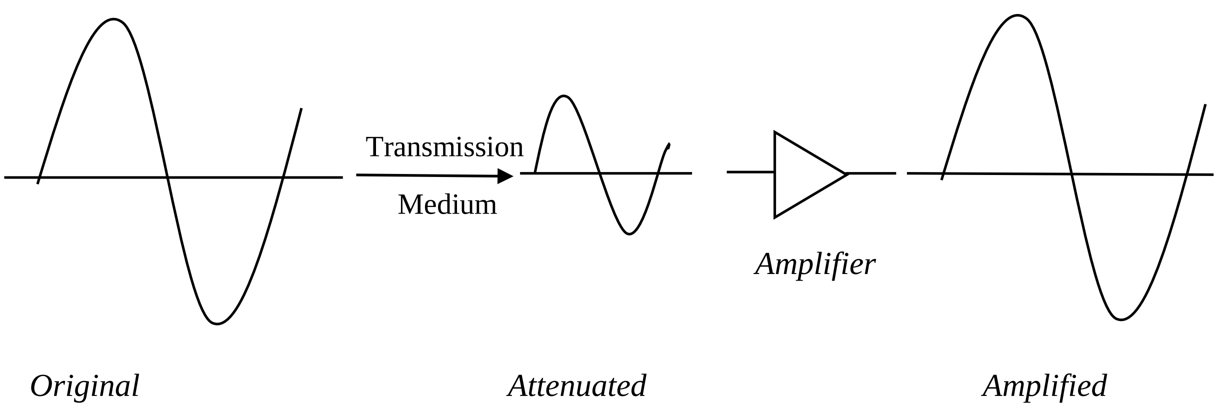
Chapter 3: Data Communication Fundamentals

Unit 2.4 Transmission Impairments (Attenuation, Distortion, Noise)

- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- Three causes of impairment are attenuation, distortion, and noise.

1. Attenuation

- When a signal travels through a medium, it loses energy overcoming the resistance of the medium.
- Amplifiers are used to compensate for this loss of energy by amplifying the signal.

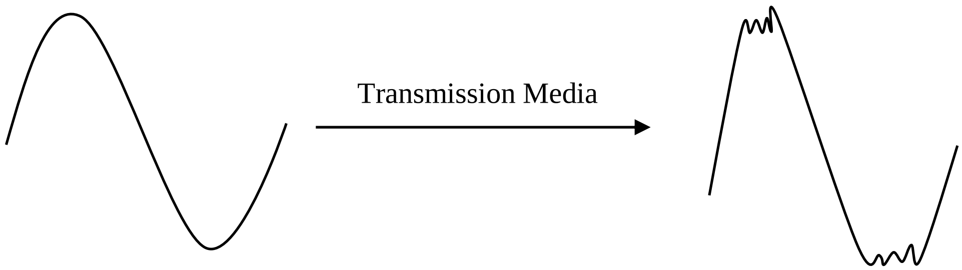


To show loss or gain;

$$\text{Loss/Gain} = 10 \log_{10}\left(\frac{P_O}{P_i}\right) \quad \text{dB}$$

2. Distortion

- It means that the signal changes its form or shape. It occurs in composite signals.
- Distortion generates additional frequencies that were not present in the original signal.



Transmitted Signal *Received Signal (Distorted)*

i. **Linear Distortion:** It occurs when a system alters the amplitude or phase of the signal's frequency components without introducing new frequencies.
ii. **Non-linear Distortion:** It occurs when a system introduces new frequencies into the original signal.

3. **Noise**

Noise refers to any unwanted or random disturbance that gets superimposed on the original signal. It is difficult to remove the unwanted noise signal.

<u>Types of noise</u>

a) Internal Noise
   i. Thermal Noise
   ii. Shot Noise
   iii. Partition Noise
   iv. Flicker Noise
   v. Transit Time Noise

b) External Noise
   i. Atmospheric Noise
   ii. Industrial Noise

a) Internal Noise

It gets generated by the electronic equipment involved in the system. Proper design of the communication system can reduce or overcome noise due to internal sources.

i. Thermal Noise

- An electrical signal is transmitted through a channel with the help of conductors. So, the electrons present in the conductors move randomly.

- The random motion of the electrons is the reason for the thermal energy received by the conductors.
- Every equipment, element and transmission medium itself contribute to thermal noise if the temperature of the element is above absolute 0. i.e. 0 K.
- This noise is also known as Johnson noise or white noise.

The noise power is,

$$P_n = KTB$$

Where, P_n= Power of noise
K= Boltzmann Constant = $1.38 * 10^{-23}$ J/K
T= Absolute Temperature
B = Bandwidth

Equivalent RMS noise voltage is,

$$E_n= \sqrt{4KTRB}; \quad R=\text{Resistance}$$

ii. Shot Noise
- It arises in active devices due to the random behavior of charge carriers.
- It is caused by random variation in the arrival of electrons or holes at the output of the amplifying device.

iii. Partition Noise
It is generated in a circuit when the current has to be divided in multiple paths. This noise is the result of random variations in divisions.

iv. Flicker Noise
It is also known as low frequency noise. Its intensity is inversely proportional to frequency. At low frequency, this type of noise appears.

v. Transit Time Noise
It is also known as high-frequency noise. It arises when a charge carriers require comparatively more time to travel from one end to another within the conductor.

b) External Noise
External noise originates outside the communication system and interferes with signal transmission.

i. Atmospheric Noise
It is caused by natural phenomena such as lightning, thunderstorms etc.

ii. Industrial Noise
It is produced by man-made sources like motors, engines, generators etc.